

Department of Artificial Intelligence & Data Science

School of Computing, FAST-NUCES

Course Title: Natural Language Processing (NLP)

Class: MS-AI/DS

Instructor: Dr. Zohair Ahmed

Total Marks: 10

Due Date: Next class. No extension. Late submissions will not be accepted.

- Viva or quiz may be conducted in the next class based on the assignment.
- Students must prepare the concepts.
- Blind copy-paste is not acceptable.

Task 1 Paper Reading and Reflection:

- **Read the first research paper (easy to read, big-picture, and motivating)**
- LeCun, Bengio, Hinton (2015), “Deep learning”, *Nature*, <https://www.nature.com/articles/nature14539>

Submit **1 page A-4** answering (Handwritten Only)

- **What is “representation learning”** and why is it the key difference from classic ML feature engineering?
- Name **3 reasons deep learning became practical** (data, compute, training tricks, etc.).
- Pick **one application** (vision/speech/NLP) and explain why a deep model fits that data structure.